IT05: Robotics and Automatic Control ロボット工学と自動制御

Quiz 5: Steady state error of PI and PD control system　PI と PD 制御系の定常誤差

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Find a steady state error of the following feedback system for each of the two different controllers. You can write your derivation down on a sheet of paper, take a photo of it, paste it to this document, and submit the document. (5 * 2 points).

下の異なる二つのコントローラに対するフィードバクシステムの定常誤差をそれぞれ求めよ．導出を紙に書き，その写真をとり，この文書にそれを貼り付けて提出しても良い．（5*2 点）．

$$\cancel{= \lim_{s\to 0} s \cdot \frac{1}{1+\frac{1}{(T_m s+1)(T_e s+1)}\cdot\left(k_p+\frac{k_i}{s}\right)} \cdot s}$$

$$\cancel{= \lim_{s\to 0} \frac{T_m T_e \cdot s^2 + (T_m+T_e)s + 1}{T_m T_e s^2 + (T_m+T_e)s + 1 + k_p + \frac{k_i}{s}}}$$

(PI) $C(s) = k_p + \frac{k_i}{s}$

$$E(s) = \frac{1}{1+\frac{1}{(T_m s+1)(T_e s+1)}\cdot\left(k_p+\frac{k_d}{s}\right)} \cdot \frac{1}{s}$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s E(s)$$

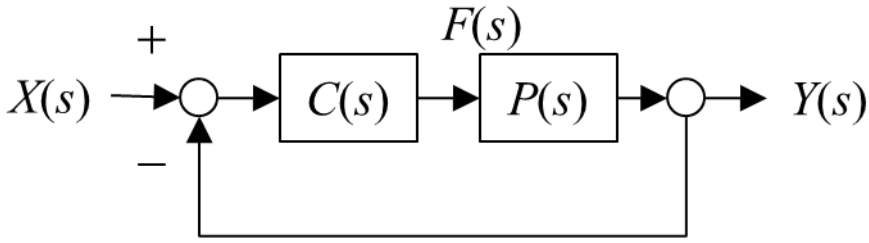
$$= \lim_{s\to 0} \frac{1}{1+\frac{1}{(T_m s+1)(T_e s+1)}\cdot\left(k_p+\frac{k_d}{s}\right)}$$

$$= \lim_{s\to 0} \frac{1}{1+\frac{1}{(T_m s+1)(T_e s+1)}\cdot\left(\frac{k_p \cdot s + k_d}{s}\right)}$$

$$= \lim_{s\to 0} \frac{(T_m s+1)(T_e s+1)\cdot s}{s(T_m s+1)(T_e s+1) + k_p \cdot s + k_d}$$

$$= \frac{0}{k_d} = 0$$

$$E(s)=X(s)-Y(s)$$



$$P(s)=\frac{1}{\left(T_m s+1\right)\left(T_e s+1\right)},\ X(s)=\frac{1}{s}$$

$$C(s)=k_p+\frac{k_i}{s},\ \text{and},\ C(s)=k_p+k_d s,$$

$$E(s)=\frac{1}{1+P(s)C(s)}X(s)=$$

$$\lim_{t\to\infty} e(t)=\lim_{s\to 0} sE(s)=$$

$$= \lim_{s \to 0} \frac{1}{1 + T_m T_e \cdot k_p s^2 + k_p (T_m + T_e) s + k_p + T_m \cdot T_e \cdot s + (T_m + T_e) + \frac{k_i}{s}}$$

(P4) $C(s) = k_p + k_d \cdot s$

$$\lim_{t \to \infty} e(t) = \lim_{s \to 0} s \cdot \frac{1}{1 + P(s) C(s)} X(s)$$

$$= \lim_{s \to 0} s \cdot \frac{1}{1 + \frac{1}{(T_m \cdot s + 1)(T_e \cdot s + 1)} \cdot (k_p + k_d \cdot s)} \cdot \frac{1}{s}$$

$$= \lim_{s \to 0} \frac{1}{1 + \frac{k_p + k_d \cdot s}{T_m T_e s^2 + (T_m + T_e) s + 1}} = \lim_{s \to 0} \frac{T_m T_e \cdot s^2 + (T_m + T_e) s + 1}{T_m T_e \cdot s^2 + (T_m + T_e) s + 1 + k_p + k_d \cdot s}$$

$$= \frac{1}{k_p + 1}$$